TU CSE

End Term Examination, December 2017,
Programming and problem solving (CS-404)

Marks :60 **Time :2 Hours**

(All Questions are compulsory)

Q1 a) Define macros in C for the following situations.

i. To calculate the simple interest on a given amount with specified interest rate and period. 2

ii. To evaluate the polynomials of the form fX^5+eX^4+bX, $fY^5+dY^3+cY^2$, fZ^5+a etc. where a, b, c, d, e, f are coefficients 4

b) int a; char c;

Using the above declarations, which of the following statements cannot be used as case label in switch statement and why. 6

i) 'A'+c   ii) 'A'+'C'   iii) 10+a

iv) 15+'a'   v) 1.5*'a'   vi) "A"+"C"

Q2. a) Draw the flow chart to sort an array of "n" elements stored as "data" using the insertion sort algorithm. 6

b) Design a composite vi editor command in symbol v, so that the last word of the current line is shifted to the beginning of the same line, irrespective of number of words in the line when v is pressed in command mode. On pressing v next time, it should affect the next line of the file. 3

c) Explain the control flow in a switch structure in C with an example and numbered arrows. Is it possible to write the first block of switch with "default" label? 3+1

d) One file with name "DIR/Data.txt" is to be handled by a C program. Read write permission is available in the directory DIR. The program opens the file, and stores a user given roll number if it is not present there. What will be the effect of two successive execution of the program, if the file is opened in "r+"? And, if the file is opened in "w+"? What is the correct solution to fulfill the requirement? 2+2+1

Q3. a) Write a program in C to find all 8-digit numbers that are perfect square. (Perfect square has integer as square root.) 4

b) Write a function in C that will check whether a integer (received as parameter) is divisible by 64 or not. You cannot use the / (div or division), % (mod) operation in it. 5

3+5
+4+5
+4
2

Q4 a) What is "register" in C? explain. 2

b) Explain about any one pre-processor directive used in C. 2

c) struct AA{ double d; char c;}; → 16

union BB{ int a; char c[11]; struct AA s;}; → 16

struct CC{int g[3]; char h[3]; struct AA m[3]; union BB n[3];}

What will be the sizes of AA, BB and CC in 32-bit system? 1+1+2

d) Assume that a program will call the function "myfunction()" based on user's choice presented through a menu. The menu will be presented any numner of times until exit option is not entered. But the computation within "myfuction()" can be used maximum 10 times. What will you do impose the mentioned restriction? 3

Q5. a) Assume that the file "data.txt" contains 90 bytes of data. What will be the output of the following program? 6

```
int main()
{ FILE *fp; fp=fopen("data.txt", "a+");
fseek(fp,20,SEEK_CUR); printf("1=%d\n",ftell());rewind(fp);
fseek(fp,-5,SEEK_CUR); printf("2=%d\n",ftell()); → 0
fseek(fp,-5,SEEK_END); printf("3=%d\n",ftell()); → 88
fseek(fp,-5,SEEK_SET); printf("4=%d\n",ftell());
fseek(fp,50,SEEK_SET); printf("5=%d\n",ftell());
fseek(fp,50,SEEK_CUR); printf("6=%d\n",ftell()); → 100
return(0); }
```

b) The function **void store(struct student arr[], int n);** will allow the user to enter information about "n" students to an array. Using the function information 10 students' information is stored in consecutive memory locations. What will you do in your programme, if you have to copy those entered information to a different memory location **without using any loop** structure. 4

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